Project Documentation

Heart Disease Analysis Dashboard using Tableau

Step 1: Project Selection

Description

The project selected for this internship is Heart Disease Analysis using Tableau. The main objective is to analyse healthcare data and identify the factors contributing to heart disease through interactive visualizations.

Output:

- Project topic finalized.
- Objectives identified.

Step 2: Dataset Collection

Description

The Heart Disease 2020 dataset was collected. The dataset contains patient demographic details, lifestyle information, and medical conditions.

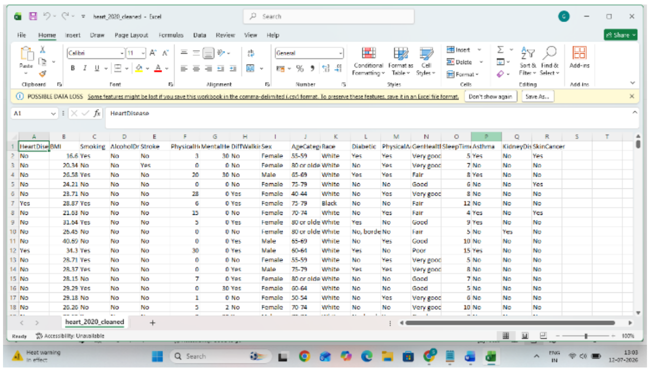
Dataset Fields

- HeartDisease
- BMI
- Smoking
- Alcohol Drinking
- Stroke
- Physical Health
- Mental Health
- Sex
- Age Category

- Race
- Diabetic
- Physical Activity
- General Health
- Sleep Time
- Asthma
- Kidney Disease
- Skin Cancer

Output:

Dataset ready for analysis.



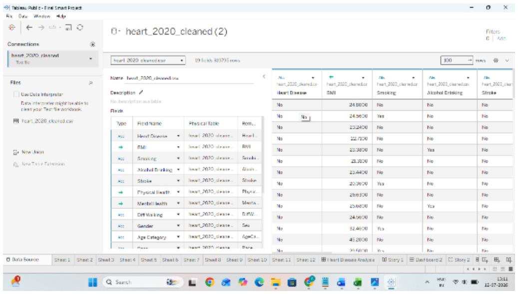
Step 3: Import Dataset into Tableau

Procedure

1. Open Tableau Desktop.
2. Click Connect.
3. Select Text File.
4. Choose the Heart Disease dataset.
5. Verify all fields.
6. Click Sheet 1.

Output

Dataset successfully imported into Tableau.



Step 4: Data Preparation

Procedure

The imported data was verified before visualization.

The following checks were performed:

- Checked Null Values
- Verified Data Types
- Renamed Columns if required
- Classified Dimensions
- Classified Measures

Dimensions

- Age Category
- Gender
- Smoking
- Diabetes
- General Health

Measures

- BMI
- Physical Health
- Mental Health

- Sleep Time

Output

Clean dataset prepared.

Step 5: Create Worksheet 1

Gender vs Heart Disease

Procedure

1. Open Worksheet.
2. Drag Heart Disease into Columns.
3. Drag Number of Records into Rows.
4. Create a Bar Chart.
5. Add Labels.

Output

Gender vs Heart Disease distribution chart created.

Step 6: Create Worksheet 2

Age vs Heart Disease

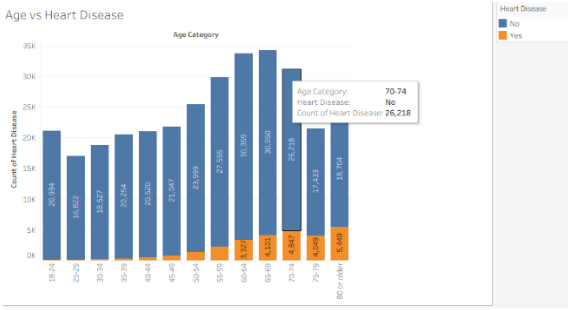
Procedure

1. Drag Age Category to Columns.
2. Drag Number of Records to Rows.
3. Filter using Heart Disease.

4. Apply colors.

Output

Age-wise heart disease analysis completed.



Step 7: Create Worksheet 3

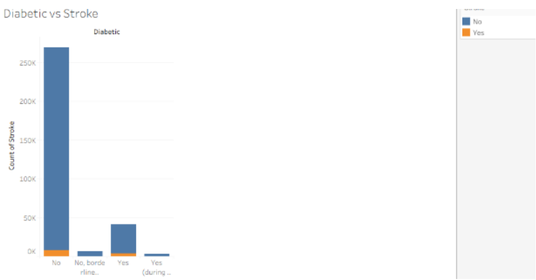
Diabetic vs Stroke

Procedure

1. Drag Sex into Columns.
2. Drag Number of Records into Rows.
3. Color by Heart Disease.
4. Add Labels.

Output

Diabetic vs Stroke chart created.



Step 8: Create Worksheet 4

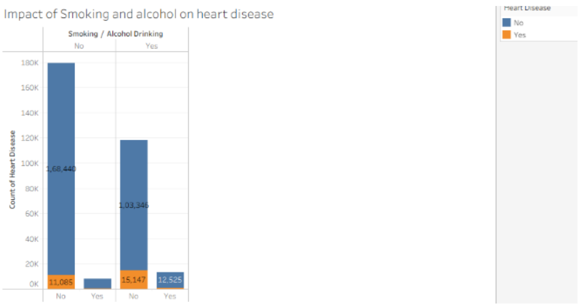
Impact of smoking and alcohol on heart disease

Procedure

1. Drag Smoking into Columns.
2. Drag Heart Disease Count into Rows.
3. Apply Filters.
4. Format the chart.

Output

Smoking and Alcohol impact analyzed.



Step 9: Create Worksheet 5

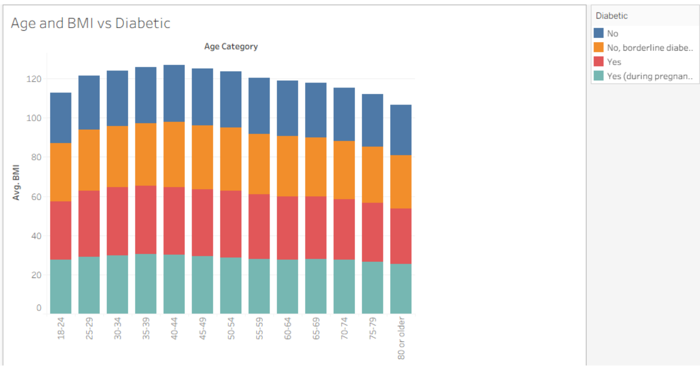
BMI Analysis

Procedure

1. Drag BMI.
2. Create Histogram.
3. Compare with Heart Disease.

Output

BMI distribution created.



Step 10: Create Worksheet 6

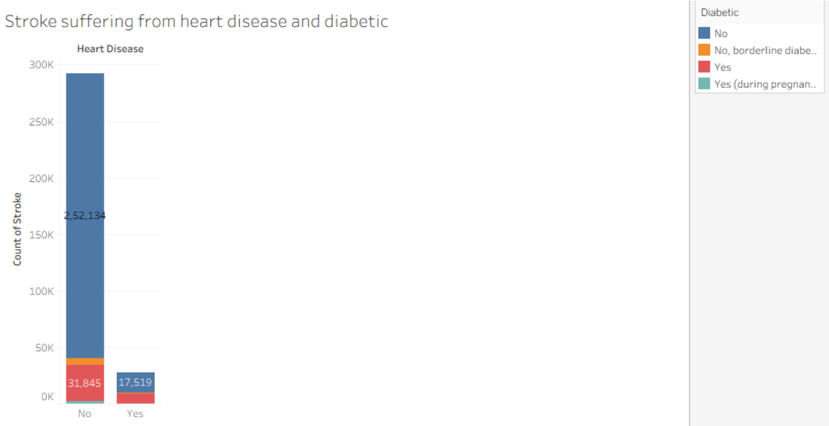
Diabetes Analysis

Procedure

1. Drag Diabetes.
2. Compare with Heart Disease.
3. Use Bar Chart.

Output

Diabetes relationship identified.



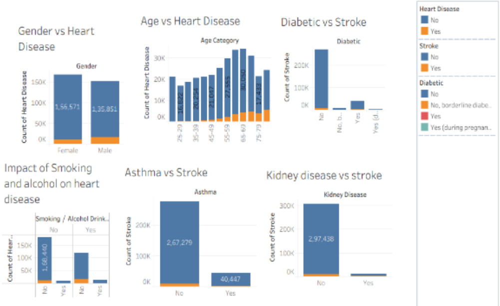
Step 11: Create Dashboard

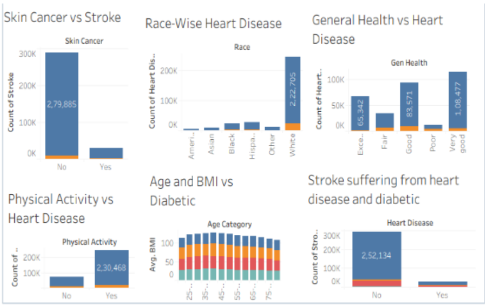
Procedure

1. Click New Dashboard.
2. Set Dashboard Size.
3. Drag all Worksheets into Dashboard.
4. Arrange Charts.
5. Add Dashboard Title.
6. Add Filters.
7. Enable Interactive Actions.

Output

Interactive dashboard completed.





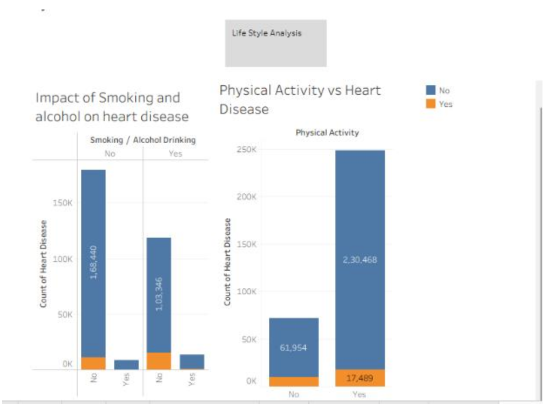
Step 12: Create Story

Procedure

1. Click New Story.
2. Add Dashboard.
3. Create Story Points.
4. Add Captions.
5. Explain Findings.

Output

Story successfully created.



Step 13: Dashboard Testing

Procedure

The dashboard was tested using:

- Filter Selection
- Highlight Actions
- Navigation
- Dynamic Updates

The dashboard responded correctly to all interactions.

Step 14: Results

The analysis identified several key findings:

- Older age groups have higher heart disease prevalence.
- Smokers are more likely to have heart disease.
- Diabetes significantly increases heart disease risk.
- Poor general health is associated with a higher incidence of heart disease.
- Physical activity is linked to a lower risk of heart disease.

Step 15: Conclusion

The Heart Disease Analysis Dashboard was successfully developed using Tableau. The dashboard transforms raw healthcare data into meaningful visual insights, allowing users to explore patterns, compare health indicators, and support informed healthcare decisions through interactive visualizations.